



Faculty of Engineering

School of Electrical Engineering and Telecommunications

ELEC 1111 – Topic 5

Inductors and RL Circuits

Dr. Inmaculada (Inma) Tomeo-Reyes

Senior Lecturer

School of Electrical Engineering and Telecommunications, UNSW

Topic 5 Content

This lecture covers:

- Inductors
- Circuit analysis with inductors
- First order circuits with resistors and inductors (RL circuits)
 - Natural response
 - Step response

Corresponds to parts of Chapter 6 and 7 of your textbook

Addressed learning outcomes:

- *LO1*: Systematically analyse DC and AC electric circuits by deriving and solving equations using circuit laws and theorems.
- *LO2*: Obtain the transient and steady state behaviour of a first order circuit.
- *LO6*: Evaluate relevant information to design simple engineering systems that use electrical circuits.

Inductors

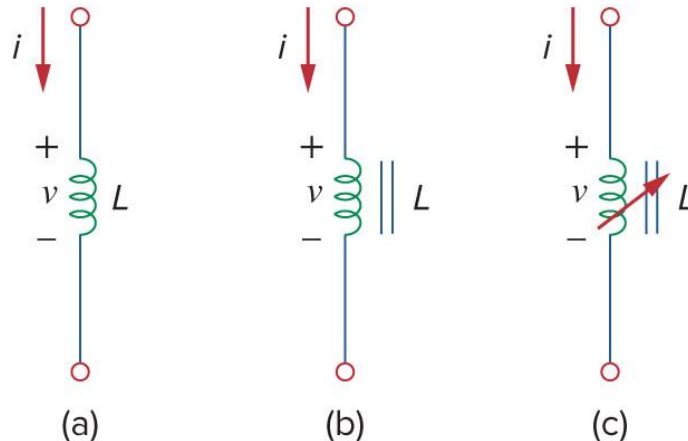
- An **inductor** is a circuit element that stores energy in its magnetic field.
 - An inductor consists of a coil of conducting wire.
 - Any conductor of electric current has inductive properties.
 - The inductive effect is typically enhanced by coiling the wire up.
- If a current is passed through an inductor, the **voltage** across it is **directly proportional** to the **time rate of change** in **current**.

$$v = L \frac{di}{dt}$$

Inductors

- **Inductance** is the property whereby an inductor exhibits opposition to the change of current flowing through it.
- The symbol for inductance is L .
- Inductance is measured in henry, H.

$$1 \text{ H} = 1 \frac{\text{V} \cdot \text{s}}{\text{A}}$$

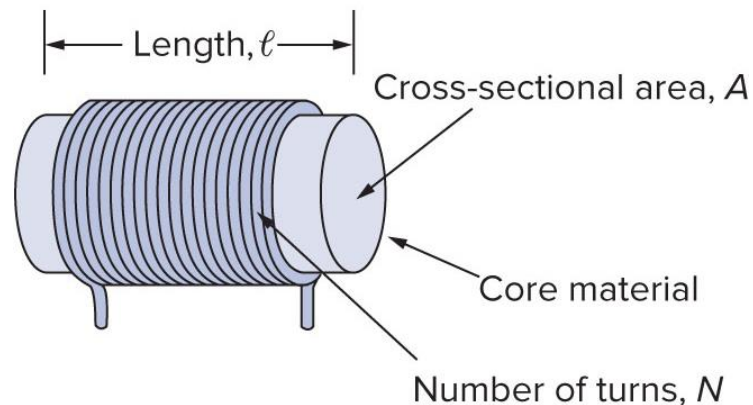


Circuit symbol for fixed inductor, (a) and (b), and variable inductor, (c).

Inductors

- Inductance depends of the **physical dimensions** and **construction** of the inductor.
- For **solenoid** inductors, the inductance is given as follows:
 - N is the **number of turns** of the wire.
 - A is the **cross-sectional** area of the core.
 - ℓ is the **length** of the core.
 - μ is the **permeability** of the core.

$$L = \frac{N^2 \mu A}{\ell}$$



Inductors

- The terms *coil* or *choke* are also used for inductors.
- Like capacitors, inductors are available in different values and types.
- They are described by their **core material**.
 - E.g. the core may be made of iron, steel, plastic, or air.
- Most inductors are rated in microhenry (μH) to tens of henrys (H).
- They are used in power supplies, transformers, radios, TVs, radars, and electric motors.



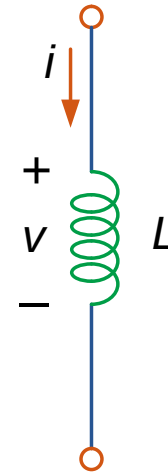
Inductors

- **Passive sign convention** applies to inductors as well.
 - If $v \times i > 0$, the inductor is being **charged** (absorbing energy).
 - If $v \times i < 0$, the inductor is **discharging** (supplying energy).
- The **inductor's voltage** v is proportional to the **rate of change** of its **current** i , with **inductance** L as the constant of proportionality, assuming **passive sign convention**.

$$v = L \frac{di}{dt}$$

- Current will then be:

$$i(t) = \frac{1}{L} \int_{t_0}^t v(\tau) d\tau + i(t_0)$$



where $i(t_0)$ is called the **initial current** or **initial conditions** at time t_0 .

Inductors

- **Instantaneous power** delivered to the inductor:

$$p = vi = Li \frac{di}{dt}$$

- The **energy** stored in the **magnetic field** that exists around the coil and wires of the can be then calculated as:

$$w(t) = \int_{t_0}^t p(\tau) d\tau = \int_{t_0}^t L \frac{di(\tau)}{d\tau} i(\tau) d\tau = L \int_{t_0}^t i(\tau) di(\tau) = L \frac{i(\tau)^2}{2} \Big|_{t_0}^t$$

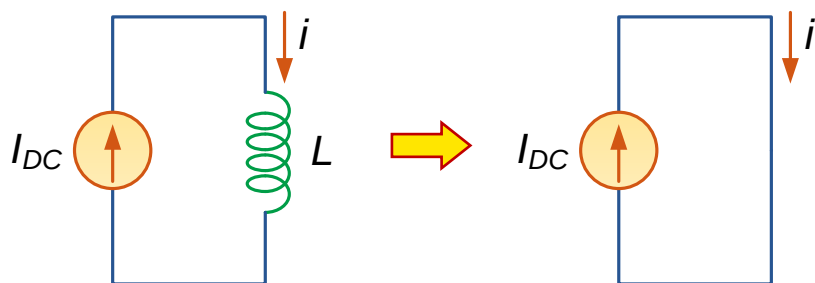
$$w(t) = \frac{1}{2} L (i(t)^2 - i(t_0)^2)$$

If $i(t_0) = 0 \rightarrow$ $w(t) = \frac{1}{2} Li(t)^2$

Properties of inductors

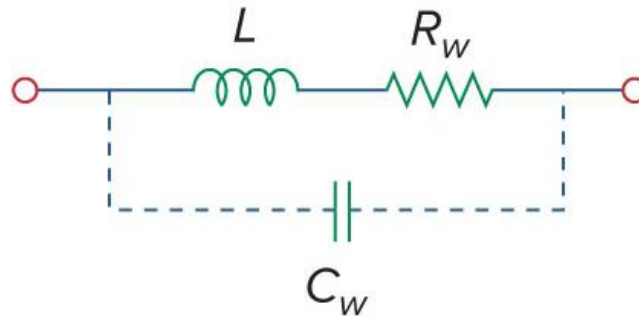
- An inductor acts like a **short circuit** to **DC current**.

$$v = L \frac{di}{dt} = 0 \text{ for constant current.}$$



Properties of inductors

- Like an ideal capacitor, an **ideal inductor** does not dissipate energy.
 - Energy is absorbed from a circuit, stored as magnetic field and then released back to the circuit.
- A **real inductor** has a very small **winding resistance** R_w in series and a very small **winding capacitance** C_w in parallel with both, leading to a slow loss of stored energy.



Inductors in series

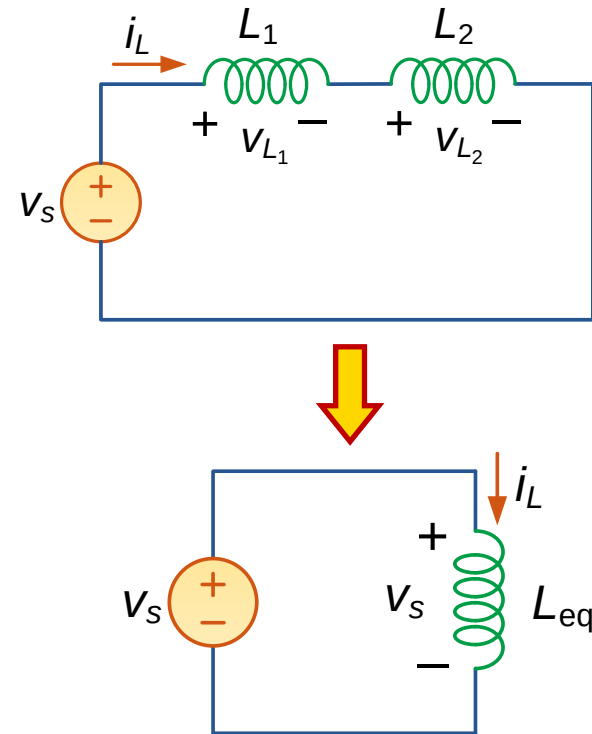
- Similar to resistors, inductors in series or parallel can be combined to simplify a circuit.
- For inductors in series, the current is the same across each inductor.
- Applying KVL and current-voltage relation $v = L \frac{di}{dt}$ for inductors:

$$v_s = v_{L_1} + v_{L_2}$$

$$v_s = L_1 \frac{di_L}{dt} + L_2 \frac{di_L}{dt}$$

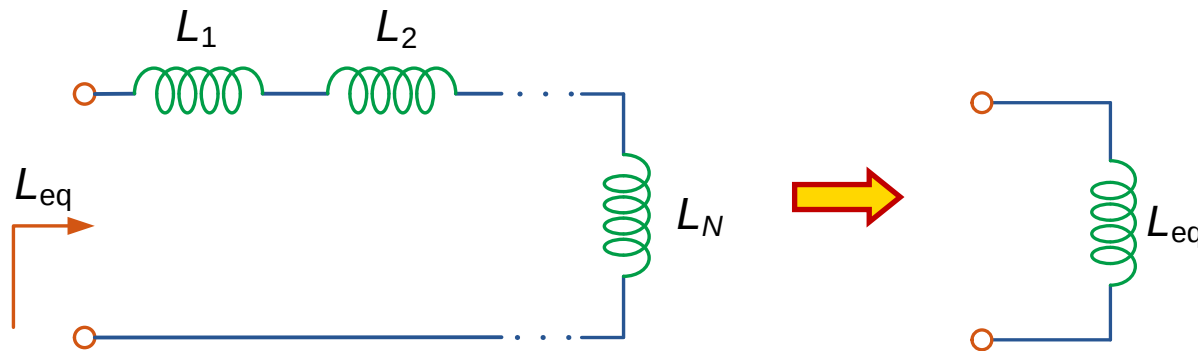
$$v_s = (L_1 + L_2) \frac{di_L}{dt} \rightarrow v_s = L_{eq} \frac{di_L}{dt}$$

$$L_{eq} = L_1 + L_2$$



Inductors in series

- The **equivalent inductance** of any number of **inductors in series** is the **sum** of the **individual inductances**.



$$L_{eq} = L_1 + L_2 + \cdots + L_N = \sum_{k=1}^N L_k$$

Inductors in parallel

- For inductors in parallel, the voltage is the same across each inductor.
- Applying KCL and voltage-current relation for inductors:

$$i_s = i_{L_1} + i_{L_2}$$

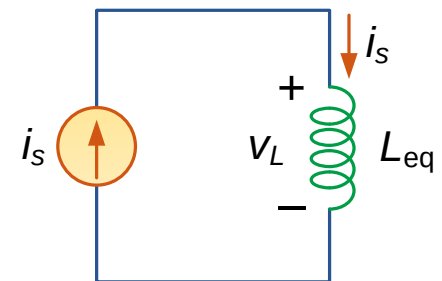
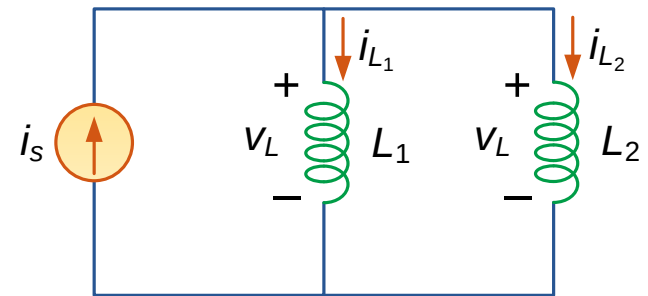
$$i_s = \frac{1}{L_1} \int_{t_0}^t v_L(\tau) d\tau + i_{L_1}(t_0) + \frac{1}{L_2} \int_{t_0}^t v_L(\tau) d\tau + i_{L_2}(t_0)$$

$$i_s = \left(\frac{1}{L_1} + \frac{1}{L_2} \right) \int_{t_0}^t v_L(\tau) d\tau + (i_{L_1}(t_0) + i_{L_2}(t_0))$$

$$\Leftrightarrow i_s = \frac{1}{L_{eq}} \int_{t_0}^t v_L(\tau) d\tau + i_{L_{eq}}(t_0)$$

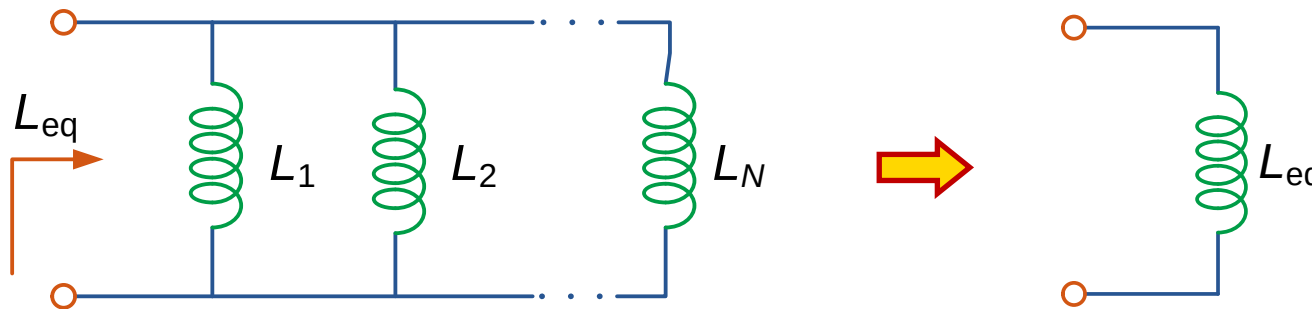
$$\boxed{\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2}} \quad \text{or} \quad \boxed{L_{eq} = \frac{L_1 L_2}{L_1 + L_2}}$$

$$\text{and } \boxed{i_{L_{eq}}(t_0) = i_{L_1}(t_0) + i_{L_2}(t_0)}$$



Inductors in parallel

- The **reciprocal** of the **equivalent inductance** of any number of **inductors in parallel** is the **sum** of the **individual reciprocal inductances**.



$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} \dots + \frac{1}{L_N}$$

or

$$L_{eq} = \frac{1}{\frac{1}{L_1} + \frac{1}{L_2} \dots + \frac{1}{L_N}}$$

Characteristics of basic elements

Important characteristics of the basic elements.[†]

Relation	Resistor (R)	Capacitor (C)	Inductor (L)
v - i :	$v = iR$	$v = \frac{1}{C} \int_{t_0}^t i(\tau) d\tau + v(t_0)$	$v = L \frac{di}{dt}$
i - v :	$i = v/R$	$i = C \frac{dv}{dt}$	$i = \frac{1}{L} \int_{t_0}^t v(\tau) d\tau + i(t_0)$
p or w :	$p = i^2 R = \frac{v^2}{R}$	$w = \frac{1}{2} C v^2$	$w = \frac{1}{2} L i^2$
Series:	$R_{eq} = R_1 + R_2$	$C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$	$L_{eq} = L_1 + L_2$
Parallel:	$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$	$C_{eq} = C_1 + C_2$	$L_{eq} = \frac{L_1 L_2}{L_1 + L_2}$
At dc:	Same	Open circuit	Short circuit

Circuit variable

that cannot

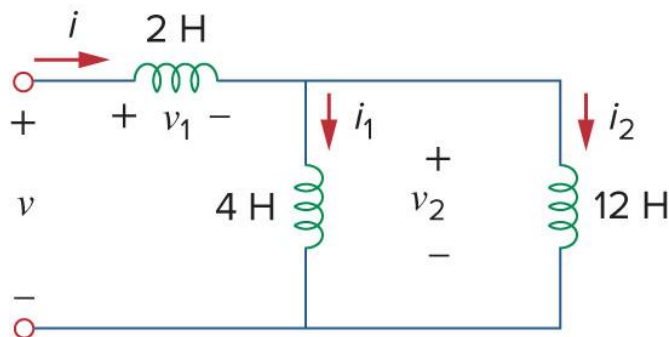
change abruptly: Not applicable v

i

[†]Passive sign convention is assumed.

Exercise 1

In the circuit below, $i(t) = 4(2 - e^{-10t})$ mA. If $i_2(0) = -1$ mA, find $i_1(0)$, $v(t)$, $v_1(t)$ and $v_2(t)$, $i_1(t)$ and $i_2(t)$.



Exercise 1

In the circuit below, $i(t) = 4(2 - e^{-10t})$ mA. If $i_2(0) = -1$ mA, find $i_1(0)$, $v(t)$, $v_1(t)$ and $v_2(t)$, $i_1(t)$ and $i_2(t)$.

- From $i(t)$, we can find $i(0) = 4 \times (2 - 1) = 4$ mA
- Using KCL: $i = i_1 + i_2$, thus:

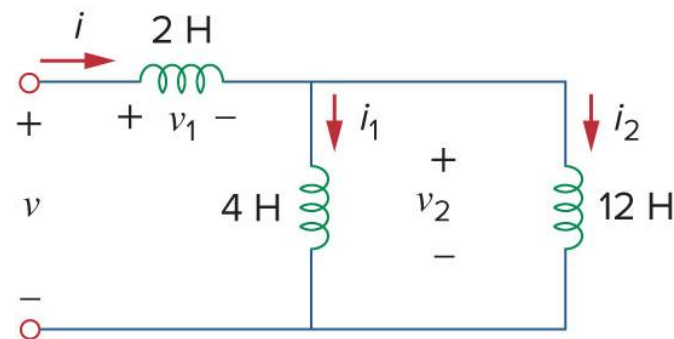
$$i_1(0) = i(0) - i_2(0) = 4 - (-1) = 5 \text{ mA}$$

- The equivalent inductance is:

$$L_{eq} = 2 + 4 \parallel 12 = 2 + 3 = 5 \text{ H}$$

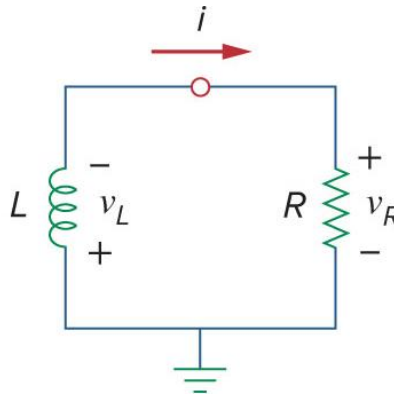
$$\text{Thus: } v(t) = L_{eq} \frac{di}{dt} = 5 \times 4 \times (-1) \times (-10)e^{-10t} = 200e^{-10t} \text{ mV}$$

- Using current $i(t)$: $v_1(t) = 2 \frac{di}{dt} = 2 \times (-4) \times (-10)e^{-10t} = 80e^{-10t} \text{ mV}$
- Using KVL: $v = v_1 + v_2$, thus: $v_2(t) = v(t) - v_1(t) = 120e^{-10t} \text{ mV}$
- Find $i_1(t)$ using $v_2(t)$: $i_1(t) = \frac{1}{4} \int_0^t v_2(\tau) d\tau + i_1(0) = \frac{120}{4} \int_0^t e^{-10\tau} d\tau + 5 = 8 - 3e^{-10t} \text{ mA}$
- Also, find $i_2(t)$ using $v_2(t)$: $i_2(t) = \frac{1}{12} \int_0^t v_2(\tau) d\tau + i_2(0) = \frac{120}{12} \int_0^t e^{-10\tau} d\tau - 1 = -e^{-10t} \text{ mA}$



Natural response of RL circuit

Let's consider a circuit with a single inductor, charged to an initial current I_0 and connected to a resistor. How will it behave?



We can use KVL to get:

$$v_L + v_R = 0$$

$$L \frac{di}{dt} + Ri = 0$$

Note: We will consider how this inductor was charged later in this lecture.

Note: A circuit characterised by a first order differential equation is called a first order circuit.

Natural response of RL circuit

Solve differential equation:

Rearrange:

$$L \frac{di}{dt} = -iR$$

$$\frac{di}{dt} = \frac{-iR}{L}$$

Separate:

$$\frac{1}{i} di = \frac{-R}{L} dt$$

Integrate:

$$\int \frac{1}{i} di = \frac{-R}{L} \int dt$$

$$\ln(i) = \frac{-R}{L} t + D$$

Solve for $v(t)$

$$i(t) = e^{-\frac{R}{L}t + D}$$

$$i(t) = e^{-\frac{R}{L}t} e^D$$

$$i(t) = Ae^{-\frac{R}{L}t}$$

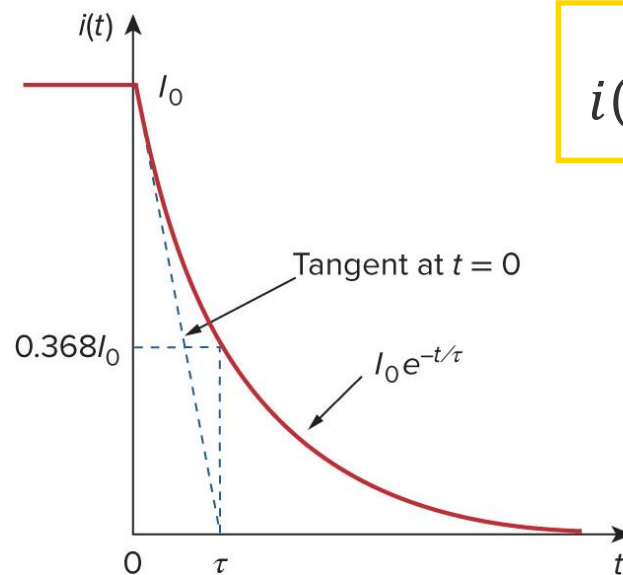
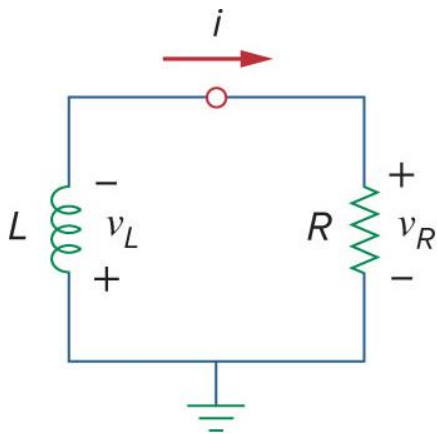
Apply initial conditions

$$i(0) = Ae^0 = A = I_0$$

$$i(t) = I_0 e^{-\frac{R}{L}t}$$

Natural response of RL circuit

- There is no need to derive the differential equation solution every time, just use the result.
- The result shows that the **current response** of the RL circuit is an **exponential decay** of the **initial current**.
- As in the case of RC circuits, the **time constant** of this circuit is the time required for the response to **decay** to $1/e$ (or **36.8%**) of its initial value or to **increase** to $1 - 1/e$ (or **63.2%**) of its final value. It is denoted by τ .



$$i(t) = I_0 e^{-\frac{R}{L}t} = I_0 e^{-\frac{t}{\tau}}$$

$$\tau = \frac{L}{R}$$

Natural response of RL circuit

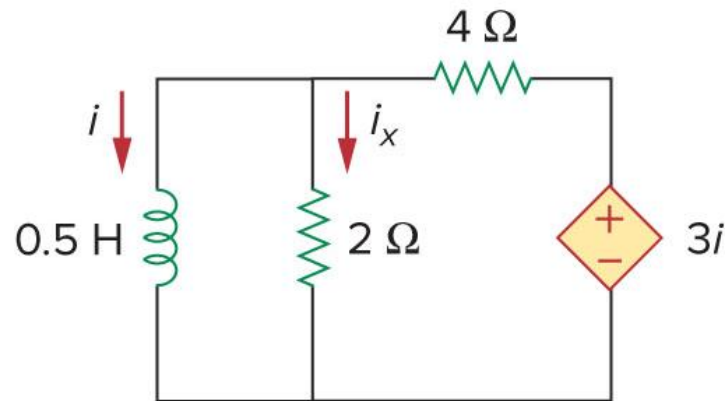
- Follow these steps to find the natural (or source-free response) of RL circuits:
 1. Find the **initial current** $i(0) = I_0$ through the inductor **before** it is connected to the resistor.
 - The inductor is assumed to be **fully charged** at the **beginning** and can be replaced with a **short circuit**.
 2. Find the **time constant** $\tau = \frac{L}{R}$.
 - If the circuit has **more than one resistor**, the resistance that we need to find in order to calculate the time constant is the equivalent resistance as seen by the terminals of the inductor, i.e. the **Thevenin equivalent resistance** $R = R_{Th}$.
 - When possible, this resistance can be obtained by simplification of series or parallel resistances.
 3. Calculate the current through the inductor as $i(t) = I_0 e^{-\frac{t}{\tau}}$.
 4. Find any other circuit variable using the inductor's current.

Note: A **switch** which **opens** or **closes** can **remove** part of the circuit or **add** something to it.

Exercise 2

Find the currents i and i_x for $t > 0$ if the initial current is $i(0) = 10$ A.

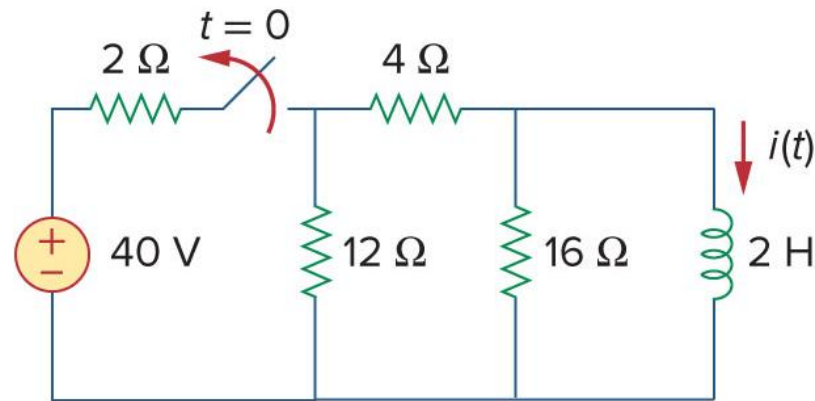
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Exercise 3

The switch in the circuit has been closed for a long time. At $t=0$, the switch is opened. Find the current $i(t)$ for $t > 0$.

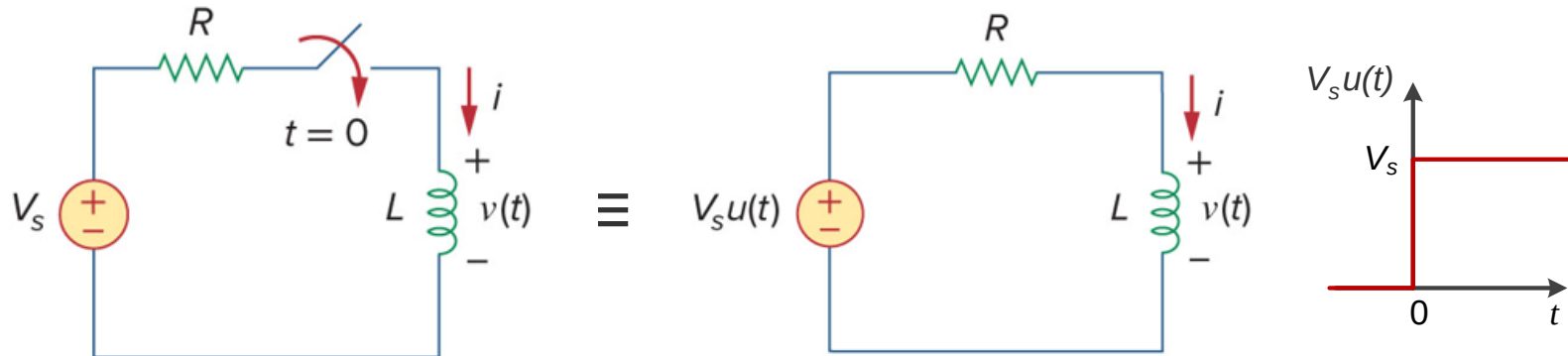
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Step response of RL circuit

- Step response is the response of the circuit due to a sudden application of a DC voltage or current.
 - It is the circuit behaviour when the **excitation/input** is the **step function**, which may be a voltage or a current source.
 - We can model this behavior with a switch opened or closed at $t = t_0$.
- Let's assume that the inductor is initially energised with a current I_0 . Since the current of the inductor cannot change instantaneously:

$$i(0^-) = i(0^+) = I_0$$



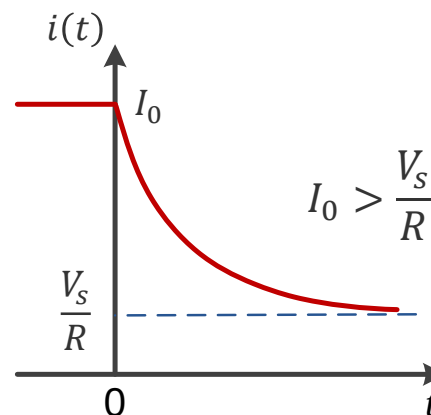
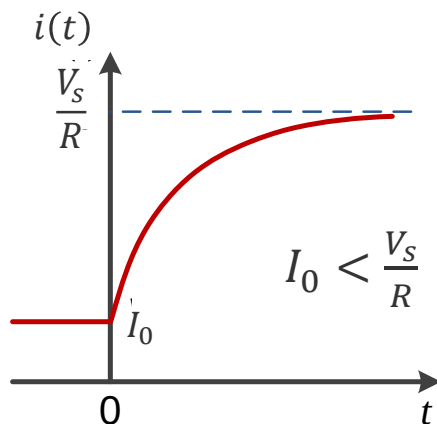
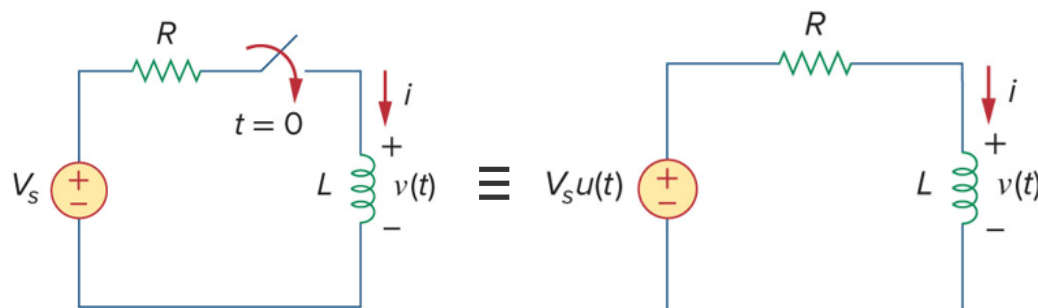
Note: $i(0^-)$ is the inductor current **just before** switching and $i(0^+)$ **just after** switching.

Step response of RL circuit

- Following an analogous process to the one shown in Topic 5 for RC circuits, we find that the **current response** of the RL circuit will change from the initial I_0 to the value of V_s/R in an **exponential manner**.
 - Depending on the value of initial conditions I_0 and voltage source V_s , the inductor can be charged or discharged.

$$i(t) = \begin{cases} I_0, & t < 0 \\ \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R}\right)e^{-\frac{t}{\tau}}, & t > 0 \end{cases}$$

$$\tau = \frac{L}{R}$$



Step response of RL circuit

- Recall that the **complete response** in **first order circuits** can be described as the sum of **transient** response i_t and **steady-state** response i_{ss} .

$$i = i_t + i_{ss}$$

- Transient response is always a decaying exponential:

$$i_t = Ae^{-\frac{t}{\tau}}, \quad \tau = \frac{L}{R}$$

- The steady-state response is obtained **a long time after** the switch is closed (about 5τ).
- Replace inductor with **short circuit** and find the current through it:

$$i_{ss} = \frac{V_s}{R}$$

- Adding the responses:

$$i = Ae^{-\frac{t}{\tau}} + \frac{V_s}{R} \quad \xrightarrow[\text{Apply initial conditions}]{i(0^-) = i(0^+) = I_0} \quad I_0 = A + \frac{V_s}{R} \rightarrow A = I_0 - \frac{V_s}{R}$$

- Therefore: $i(t) = \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R}\right)e^{-\frac{t}{\tau}}$

Step response of RL circuit

- More specifically:

1. First stage of steady-state ($t < 0$):

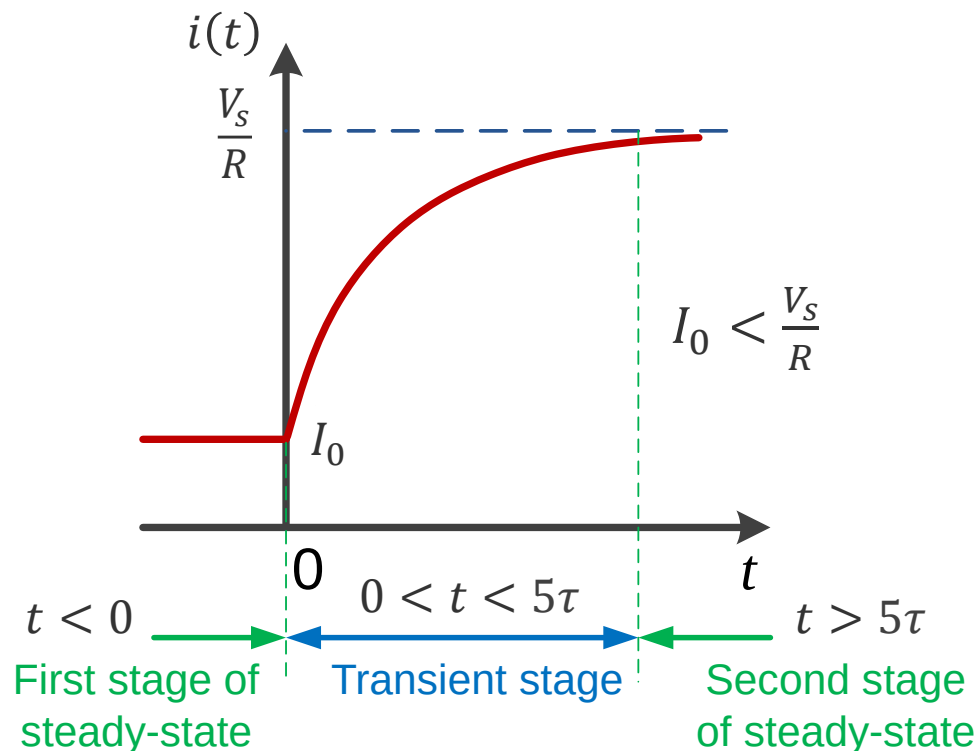
There has been **no change** in the circuit for a **long time** and the inductor is a **short circuit** with $i(0) = I_0$.

2. Transient stage ($0 < t < 5\tau$):

The inductor's voltage changes **exponentially** with $\tau = \frac{L}{R}$.

3. Second stage of steady-state ($t > 5\tau$):

The inductor's current reaches its **final value** or **steady-state value** and becomes **short circuit** again with $i(t) = \frac{V_s}{R}$ when $t \rightarrow \infty$ or $i(\infty) = \frac{V_s}{R}$.



$$\frac{V_s}{R} + \left(I_0 - \frac{V_s}{R}\right) e^{-\frac{t}{\tau}}, \quad t > 0$$

$$i(t) = i(\infty) + [i(0) - i(\infty)] e^{-\frac{t}{\tau}}, \quad t > 0$$

Step response of RL circuit

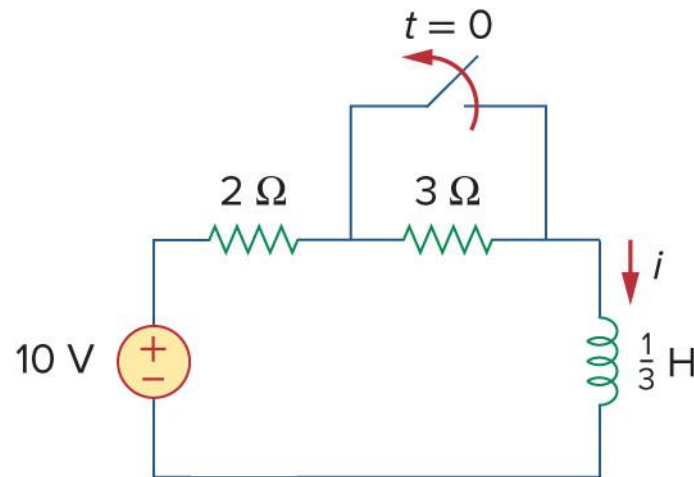
- Follow these steps to find step response of RL circuits:
 1. Find the **initial current** $i(0)$ at $t = 0$ through the inductor **before any changes** in the circuit ($t < 0$).
 - The inductor is assumed to be a **short circuit**.
 2. Find the **final current** $i(\infty)$ at $t \rightarrow \infty$ through the inductor **after the changes** in the circuit ($t \geq 0$).
 - The inductor is assumed to be a **short circuit**.
 3. Find the **time constant** $\tau = \frac{L}{R_{Th_\infty}}$ **after the changes** in the circuit.
 - R_{Th_∞} is Thevenin equivalent resistance **after the changes** ($t \geq 0$).
 4. Calculate the current through the inductor as:
$$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-\frac{t}{\tau}}$$
 5. Find any other circuit variable using the inductor's current.

Note: A **switch** which **opens** or **closes** can **remove** part of the circuit or **add** something to it.

Exercise 4

Find $i(t)$ in the circuit for $t > 0$ assuming that the switch has been closed for a long time.

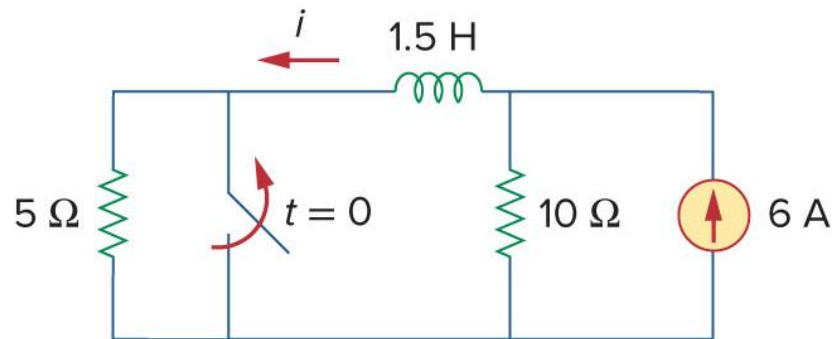
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Exercise 5

Find $i(t)$ in the circuit for $t > 0$ assuming that the switch has been closed for a long time.

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Questions?

